SYSTEM ARCHITECTURE · VERIFIED AGAINST PRODUCTION

How a Form 4 becomes a signal

LookInsight reads every SEC Form 4 filed each trading day, keeps only genuine open-market purchases, and emits a signal when two or more insiders at the same midcap company buy inside a thirty-day window. Everything below is drawn from the running system, not from a design document.

MATURED SIGNALS	HIT RATE	AVG RETURN	VS SPY	ALPHA
193	65.3%	+13.52%	+6.20%	+7.32pp

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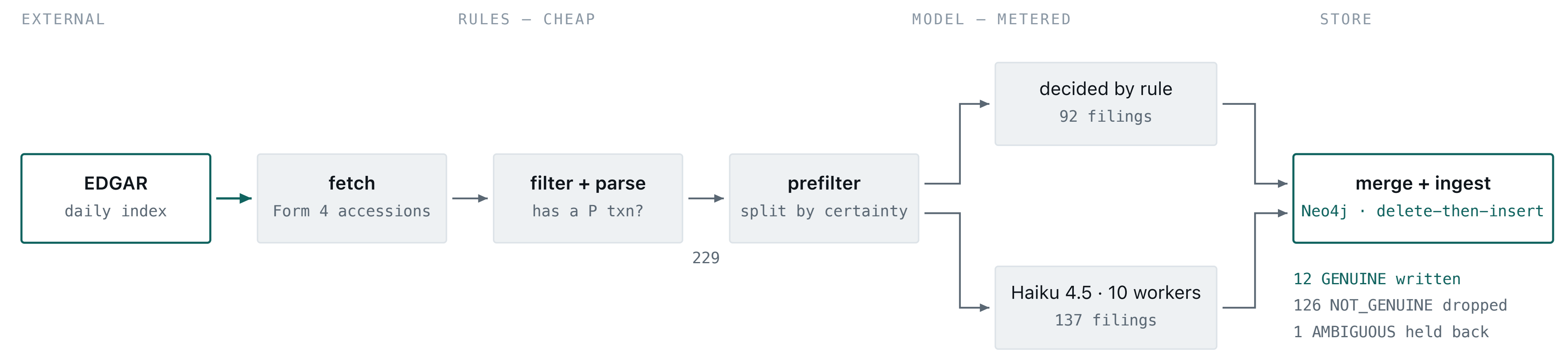
01 · INGESTION

The daily pipeline

One command per trading day walks seven steps. The expensive step — asking a language model whether a purchase is genuine — runs last and only on filings that the cheap rules could not decide, which is what keeps a day's classification cost in the cents rather than the dollars.

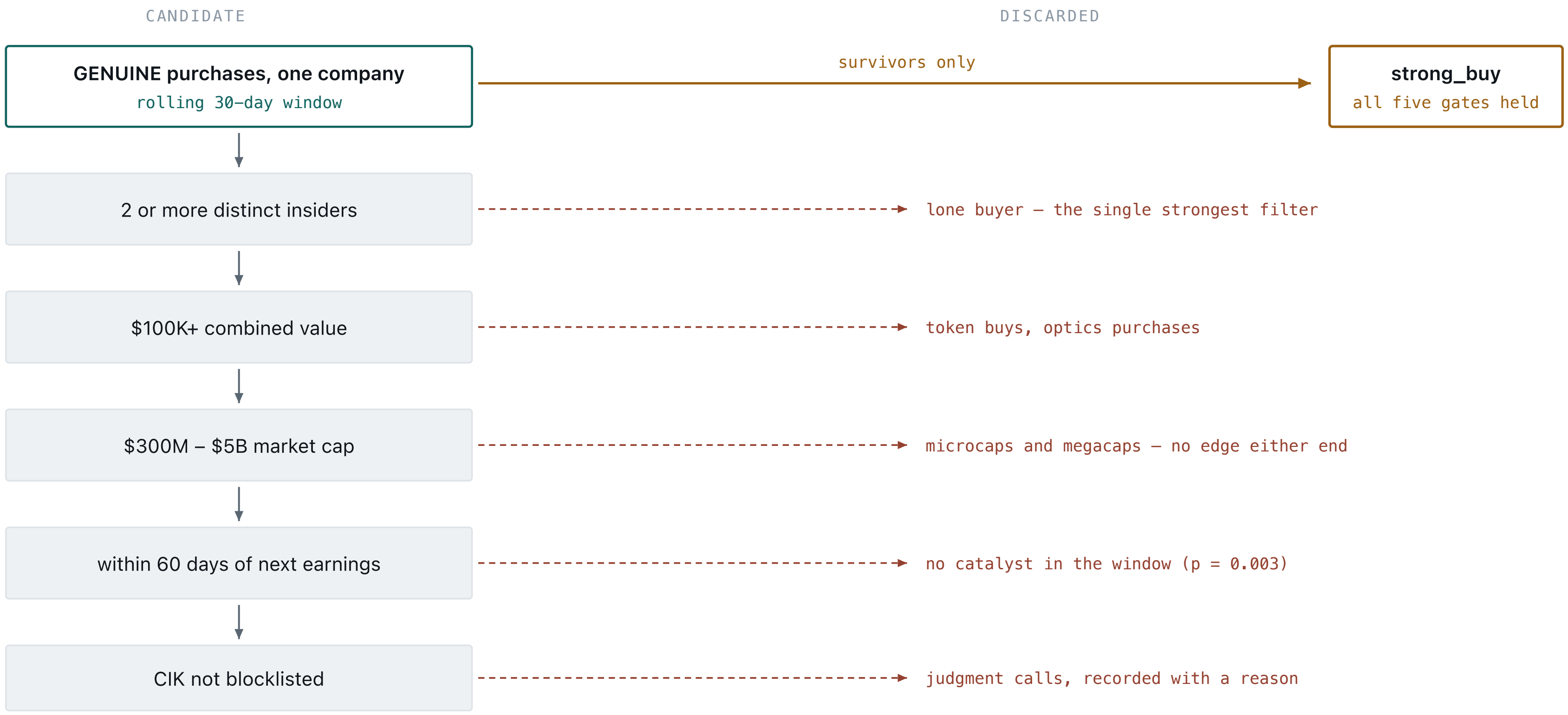


Counts are the real Aug 12–13 2026 run: 229 filings carrying a purchase code reached the prefilter, 92 were settled by rule and 137 went to the model, and 12 transactions survived as genuine open-market buys. Roughly 95% of what EDGAR publishes is discarded before it can ever become a signal.

02 · SIGNAL FORMATION

Five gates, and what each one throws away

A genuine purchase is not a signal. Five conditions have to hold at once, and the interesting part of the design is the reject path — most clusters die at the second or third gate. The thirty-day window is frozen at the moment the cluster forms: a sixth insider buying on day 40 does not retroactively join it, because at decision time you would not have known.

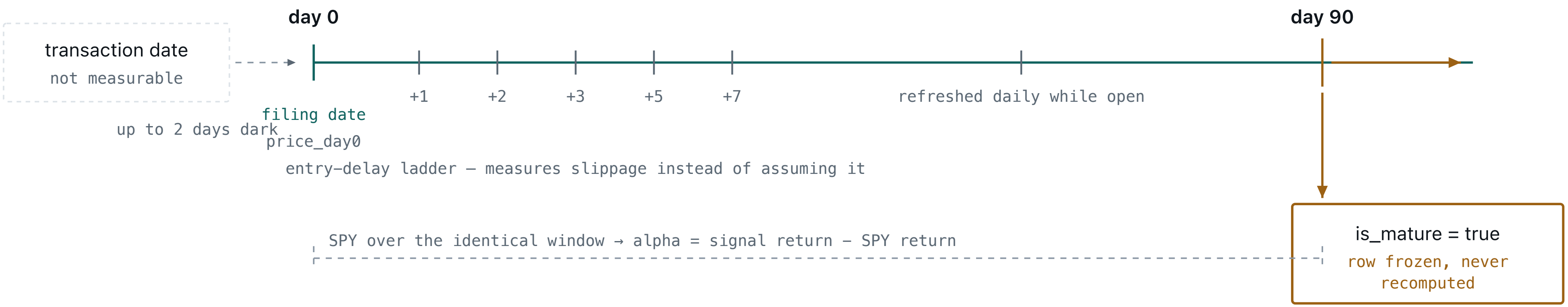


The gates run in this order because each is cheaper than the next. Two names cleared all five in the last month; on Aug 12–13 nothing did, which is the normal state — the system is designed to emit rarely.

03 · MEASUREMENT

Where the returns are measured from

This is the part worth being pedantic about. Returns run from the *filing* date, not the transaction date, because the transaction is invisible to the public until the Form 4 lands — measuring from the trade date would credit the strategy with days no one could have acted on. Six entry delays are stored so slippage can be examined rather than assumed, and at day 90 the row is marked against SPY over the identical window and then frozen.

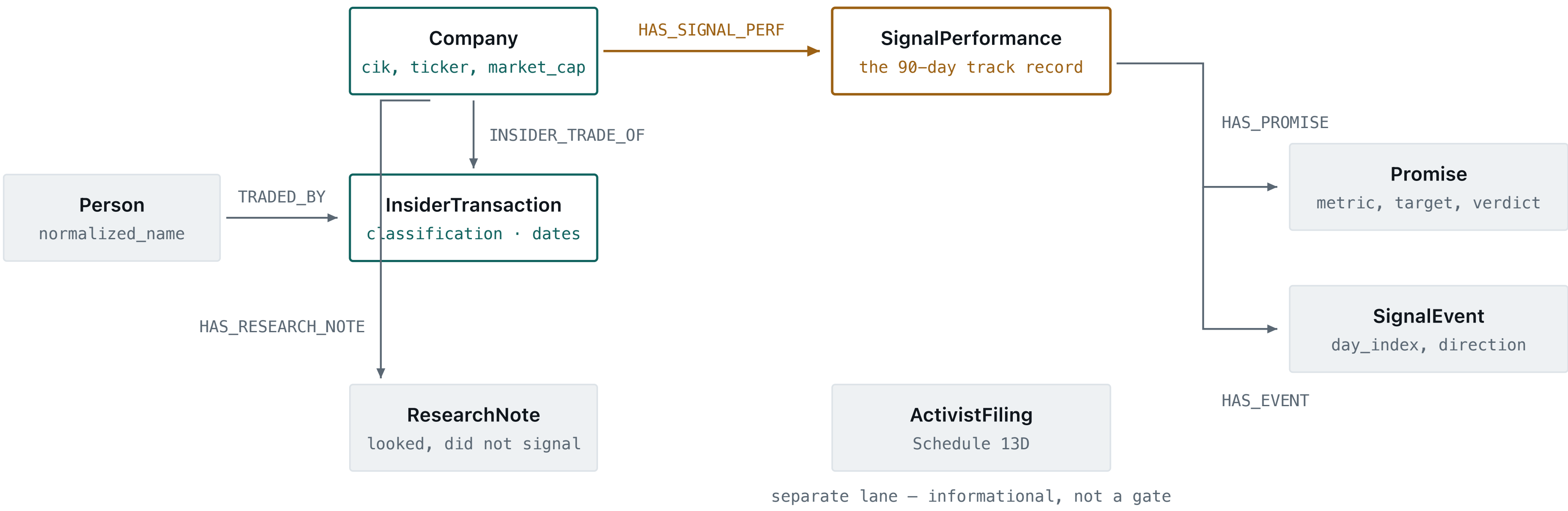


Maturation happens only after day 90 has fully passed, never on day 90 itself, and a matured row is immutable — new rules apply forward only. That freeze is what stops the published track record from drifting every time the methodology improves.

04 · STORAGE

Why a graph, and what hangs off what

The questions this product answers are traversals — which people bought at which companies, which of those clusters became signals, what each signal's management promised and whether they delivered. In a relational schema those are joins; here they are edges.

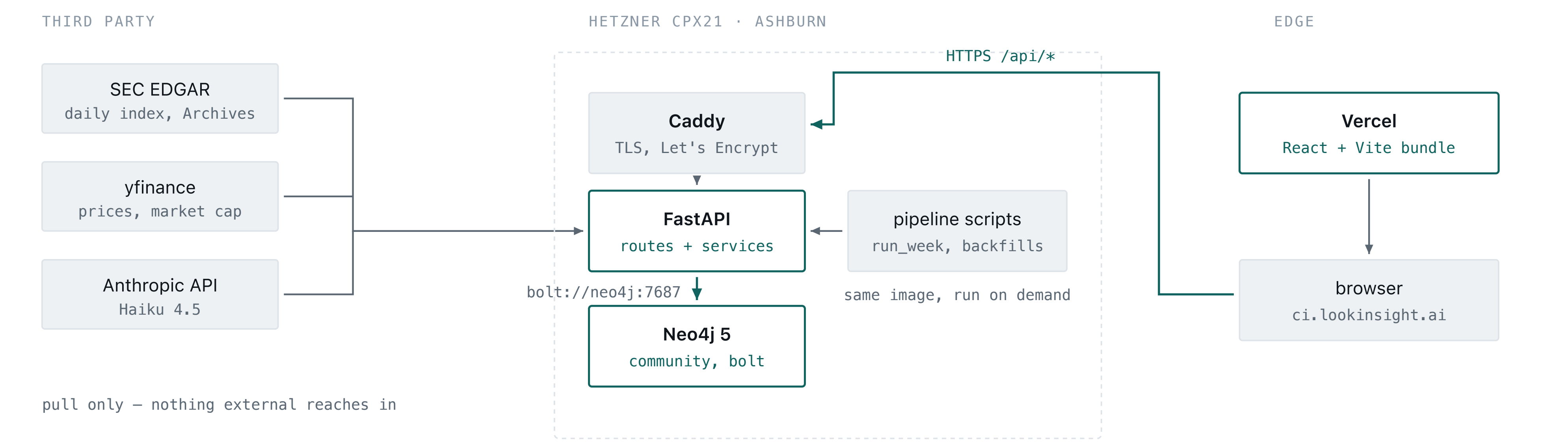


18,174 InsiderTransaction nodes as of Aug 13 2026, 204 SignalPerformance rows, and 52 Promise nodes across the ten open signals. Promise and SignalEvent are the newest layer: they record what management committed to on the call that preceded the insider buying, and whether the next call delivered it.

05 · RUNTIME

What runs where

The whole backend is three containers on one Hetzner box in Ashburn. The database moved off managed hosting in May 2026 and the monthly bill went with it; the frontend is a static bundle on Vercel that talks to the box over TLS terminated by Caddy.



The pipeline scripts run inside the same backend image rather than as a separate service, so an ingest run shares the application's code and database credentials exactly. Nothing external can reach in; every third-party dependency is pulled on a schedule the operator triggers.

06 · STRUCTURE

The dependency rule

Four layers, enforced rather than documented. The rule that matters is the one about scanners: a scanner may not import from the API layer, and ingestion may not import from services. Break either and detection logic starts depending on how a route happens to be shaped, which is how a data pipeline quietly becomes untestable.

LAYER	HOLDS	MAY DEPEND ON
`data`	Neo4j client, models	shared — reachable from everywhere
`ingestion`	EDGAR fetchers, Form 4 parsers	data only — never services or API
`domain`	services, scanners, signal logic	data + ingestion
`api`	FastAPI routes	domain + data
`delivery`	frontend, CSV exports	api

Two structural budgets sit alongside it: zero dependency cycles, and no file over 500 lines. Both are checked on every pass rather than trusted.

Drawn from the running system on 2026-08-14. Cohort statistics queried live: 193 matured strong_buy signals, 65.3% hit rate, +13.52% average 90-day return against +6.20% for SPY over matched windows. Graph current through filing date 2026-08-13.